ECOVISION ACADEMY

COMPLETE SQL MENTORSHIP PROGRAMME

Course Fee:	Total Duration:	Level:
₹399 Only	30 Hours	Beginner to Advanced

✓ **Outcome:** Job-ready SQL + Analytical Thinking + Interview Confidence.

MySQL™

SQL SQL

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SECTION 1: Foundations of Data & SQL

Duration: 1.5 Hours

- Understanding Data in the Real World
- Difference between Data, Information & Knowledge
- What is a Database Management System (DBMS)
- Types of Databases: Relational vs Non-Relational
- Introduction to SQL (Structured Query Language)
- Role of SQL in Data Science, Analytics, Finance & IT
- Overview of SQL Standards (MySQL, PostgreSQL, SQL Server, Oracle)
- Course Roadmap & Expectations

SECTION 2: SQL Environment Setup & Tools

Duration: 1.5 Hours

- Installing PostgreSQL Database
- Installing and Understanding pgAdmin
- Creating and Managing Databases
- Database Schema Concept
- Writing and Executing SQL Queries
- Understanding Query Execution Flow
- Best Practices for Beginners

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SECTION 3: SQL SYNTAX & SELECT STATEMENT

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SECTION 3: SQL Syntax & SELECT Statement

Duration: 2 Hours

- SQL Syntax Rules & Formatting
- SELECT Statement in Detail
- Selecting Columns & Expressions
- Using DISTINCT
- Column Aliasing
- ORDER BY for Sorting Data
- LIMIT and OFFSET for Result Control
- Practical Query Writing Session

SECTION 4: Data Filtering Using WHERE Clause

Duration: 2 Hours

- Logical Flow of SQL Queries
- WHERE Clause Fundamentals
- Comparison Operators
- Logical Operators (AND, OR, NOT)
- BETWEEN & IN Operators
- Pattern Matching using LIKE
- Handling NULL Values
- Filtering Real-Life Business Data

MASTERING SQL SKILLS

SECTION 5: Aggregate Functions & Grouping

Duration: 2 Hours

- Understanding Aggregation
- COUNT, SUM, AVG
- MIN & MAX
- GROUP BY Clause Explained
- HAVING Clause
- Difference Between WHERE & HAVING
- Group-wise Analysis
- Business-Oriented Examples

SECTION 6: SQL Data Types & Constraints

Duration: 1.5 Hours

- Numeric, Character & Date Data Types
- Choosing Correct Data Types
- Primary Key Concept
- Foreign Key Relationships
- NOT NULL, UNIQUE, DEFAULT
- CHECK Constraint
- Data Integrity Rules

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SECTION 7 & 8:
TABLE CREATION & DATA MANIPULATION | UNDERSTANDING TABLE RELATIONSHIPS

SECTION 7: TABLE CREATION & DATA MANIPULATION

Duration: 2 Hours

- CREATE TABLE in Detail
- INSERTING Records
- Updating Existing Records
- Deleting Records Safely
- TRUNCATE vs DELETE
- DROP TABLE
- ALTER TABLE Operations
- Data Modification Best Practices

SECTION 8: UNDERSTANDING TABLE RELATIONSHIPS

Duration: 1.5 Hours

- One-to-One Relationship
- One-to-Many Relationship
- Many-to-Many Relationship
- Concept of Referential Integrity
- Practical Relationship Design

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SECTION 9

SQL JOINS — THE POWER OF DATA CONNECTIONS

Duration: 2 Hours

Because real data never lives in a single table.

- Why **JOINS** are the backbone of relational databases
- **INNER JOIN** — extracting perfectly matched records
- **LEFT & RIGHT JOIN** — handling partial data smartly
- **FULL JOIN** — complete data visibility
- Writing accurate **JOIN** conditions
- Visual explanation using tables & diagrams
- Hands-on practice with real-world datasets

SKILL GAINED:
Ability to merge multiple tables and extract meaningful insights like a data professional.

SECTION 10: ADVANCED SQL JOIN OPERATIONS

Duration: 1.5 Hours

Move from basic queries to performance-driven SQL.

- Joining multiple tables seamlessly
- **SELF JOIN** — mastering hierarchical data
- Combining **JOINS** with aggregate functions
- Query optimization & performance tips
- Avoiding common **JOIN** mistakes used by beginners

SKILL GAINED:
Write optimized, efficient JOIN queries used in analytics, reporting & databases.

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SECTION 11

SUBQUERIES & NESTED QUERIES

Duration: 1.5 Hours

- What are Subqueries
- Subqueries in WHERE Clause
- Subqueries in SELECT Clause
- Subqueries in FROM Clause
- Correlated Subqueries
- Practical Problem Solving

12 SECTION 12: CONDITIONAL LOGIC IN SQL

Duration: 1 Hour

- CASE WHEN Statement
- CASE with Aggregate Functions
- Nested CASE Statements

AZ SECTION 13: STRINGS & TEXT FUNCTIONS

Duration: 1 Hour

- Text Manipulation Concepts
- UPPER, LOWER
- LENGTH & POSITION
- SUBSTRING
- CONCAT & TRIM
- Cleaning Text Data

SKILL GAINED:

Write optimized, efficient 101N queries used in analytics, reporting & databases.

SQL FUNCTIONS & VIEWS

SECTION 14: NUMERIC & MATHEMATICAL FUNCTIONS

Duration: 1 Hour

- ROUND & TRUNCATE
- FLOOR & CEILING
- MOD Function
- POWER & RANDOM
- Numeric Data Handling

SECTION 15: DATE & TIME FUNCTIONS

Duration: 1.5 Hours

- Date & Timestamp Data Types
- CURRENT_DATE & NOW
- Extracting Date Parts
- Date Arithmetic
- Interval Calculations

SECTION 16: VIEWS & VIRTUAL TABLES

Duration: 1 Hour

- Creating Views
- Updating Views
- Dropping Views
- Use Cases in Reporting

ADVANCED SQL CONCEPTS

SECTION 17: INDEXING & PERFORMANCE BASICS

Duration: 1 Hour

- What is Indexing
- Types of Indexes
- Creating Indexes
- How Indexes Improve Performance
- When NOT to Use Indexes

SECTION 18: TRANSACTIONS & DATA SAFETY

Duration: 1 Hour

- ACID Properties
- BEGIN Transaction
- COMMIT
- ROLLBACK
- SAVEPOINT
- Safe Data Handling

SECTION 19: WINDOW FUNCTIONS (ADVANCED SQLs)

Duration: 1.5 Hours

- Introduction to Window Functions
- ROW_NUMBER
- RANK & DENSE_RANK
- PARTITION BY
- Running Totals & Moving Averages

ADVANCED SQL TECHNIQUES

SECTION 20: STORED PROCEDURES & FUNCTIONS

Duration: 1.5 Hours

- Introduction to Stored Procedures
- Creating User-Defined Functions
- Parameters & Return Values
- Practical Automation Examples

SECTION 21: DATABASE NORMALIZATION

Duration: 1 Hour

- Importance of Database Design
- First Normal Form (1NF)
- Second Normal Form (2NF)
- Third Normal Form (3NF)
- Practical Table Design

SECTION 22: SQL FOR DATA ANALYSIS

Duration: 1.5 Hours

- Analytical Thinking with SQL
- KPI Calculation Methods
- Trend & Growth Analysis
- Business Reporting Queries

ADVANCED SQL PRACTICE

SECTION 23: WORKING WITH REAL-WORLD DATA

Duration: 1.5 Hours

- Importing External Data
- Cleaning Dirty Data
- Handling Missing Values
- Industry-Style Datasets

SECTION 24: SQL MINI PROJECTS

Duration: 2 Hours

- Student Management System
- Sales & Revenue Analysis
- Employee Database Project
- Hands on Query Practice

SECTION 25: SQL FOR INTERVIEWS & EXAMS

Duration: 1 Hour

- Common Interview Questions
- Query-Based Tests
- Conceptual Questions
- Mistakes to Avoid

FINAL SQL STAGES:
EXCEL, PRACTICE & CERTIFICATION

SECTION 26: SQL WITH EXCEL & BI TOOLS

Duration: 1 Hour

- SQL + Excel Workflow
- Exporting SQL Data
- Introduction to SQL for BI Tools

SECTION 27: REVISION & PRACTICE

Duration: 1 Hour

- Important Commands Recap
- Syntax Revision
- Error Debugging
- Practice Sets

SECTION 28: FINAL ASSESSMENT & CERTIFICATION

Duration: 1 Hour

- Final SQL Test
- Project Evaluation
- Certification Criteria
- Career Roadmap